

Active-Directory-Endpoint-Integration-Lab

End-to-end deployment and integration of a Windows 11 Enterprise workstation into an isolated Active Directory domain environment using an internal virtual network topology.

Project Overview

This project details the end-to-end engineering, provisioning, and integration of a secure, isolated enterprise network infrastructure. Using Oracle VirtualBox, a primary Windows Server Domain Controller was deployed alongside a Windows 11 Enterprise client node. The environment was engineered using a strictly non-routable internal network topology to eliminate external attack vectors and simulate a secure corporate sandbox.

Architectural Highlights (Recruiter Summary)

- **Network Isolation:** Configured an isolated internal network switch (`intnet`) to enforce strict local routing boundaries, demonstrating a strong understanding of sandbox security.
- **Identity & Access Management (IAM):** Provisioned a localized Active Directory forest (`mydomain.local`) utilizing proper static IPv4 schemas and integrated DNS zone hierarchies.
- **System Optimization & Troubleshooting:** Resolved interface routing loops caused by legacy NAT bindings and programmatically adjusted host firewalls to allow deterministic enterprise traffic.

Phase 1: Domain Controller Provisioning & Core Networking

The initial phase focused on deploying the foundational infrastructure host, establishing static addressing bounds, and configuring core Active Directory roles.

Step 1: Base Virtual Appliance Network Configuration

[Image Linked: screenshots/screenshot_1.png]

Technical Context: Initialized the virtualization layer network parameters for the primary host node, binding interface controls to structural baseline segments.

Step 2: Active Directory Domain Services (AD DS) Initialization

[Image Linked: screenshots/screenshot_2.png]

Technical Context: Successfully provisioned the AD DS and DNS roles via Server Manager, preparing the server for local forest promotion.

Phase 2: Client Virtual Appliance Layer Staging

This phase documents the provisioning of the secondary Windows 11 workstation, configuring virtual storage allocations, and setting up the guest OS environment.

Step 3: Virtual Disk and Hardware Allocation

[Image Linked: screenshots/screenshot_3.png]

Technical Context: Allocated storage blocks and virtual disk layouts (`.vdi`) within the hypervisor layer to sustain the client node profile.

Step 4: Hypervisor Interface Hardening

[Image Linked: screenshots/screenshot_4.png]

Technical Context: Configured physical-to-virtual network interface card (NIC) bindings on the workstation, transitioning infrastructure components away from default dynamic pools.

Step 5: Storage Volumetrics & Target Drive Partitioning

[Image Linked: screenshots/screenshot_5.png]

Technical Context: Structured unallocated storage blocks during the Windows 11 installation sequence to establish localized root system directories.

Step 6: Local Security Account Manager (SAM) Initial Provisioning

[Image Linked: screenshots/screenshot_6.png]

Technical Context: Provisioned the foundational local administrative account profile (`ClientAdmin`) during the operating system Out-Of-Box Experience (OOBE).

Step 7: Hypervisor Guest Addition Subsystem Deployment

[Image Linked: screenshots/screenshot_7.png]

Technical Context: Mounted and compiled guest virtualization subsystem drivers to enable dynamic hardware rendering and cross-boundary kernel utilities.

Phase 3: Infrastructure Alignment & Network Optimization

This phase focuses on standardizing host configurations, modifying IP allocation layers, and resolving critical network routing anomalies across the internal switch.

Step 8: Hostname Standardization

[Image Linked: screenshots/screenshot_8.png]

Technical Context: Enforced systematic naming conventions by changing the client machine's generic computer name to a structured asset tracking ID (`W11-CLIENT-01`).

Step 9: Static IPv4 Schema & DNS Pointer Mapping

[Image Linked: screenshots/screenshot_9.png]

Technical Context: Configured static IPv4 addressing (`172.16.0.100/24`) on the workstation, setting the primary DNS pointer directly to the Domain Controller interface (`172.16.0.10`).

Step 10: Connectivity Validation & Inter-VM Routing Fix

[Image Linked: screenshots/screenshot_10.png]

Technical Context: Successfully resolved routing conflicts by disconnecting legacy NAT adapters, resulting in full Layer 3 connectivity across the `intnet` vSwitch with 0% packet loss.

Phase 4: Active Directory Forest Integration

The final phase details the cryptographic domain-join sequence, administrative verification, and profile container synchronization on the target network.

Step 11: Domain Directory Authentication Challenge

[Image Linked: screenshots/screenshot_11.png]

Technical Context: Triggered the interactive domain-join challenge, successfully prompting the client machine to locate the `mydomain.local` directory zone via authoritative DNS lookups.

Step 12: Successful Domain Admission

[Image Linked: screenshots/screenshot_12.png]

Technical Context: Successfully authenticated administrative credentials, generating an active object reference in the domain controller database and admitting the client machine to the forest.

Step 13: Centralized Identity Session Initiation

[Image Linked: screenshots/image_148420.jpg]

Technical Context: Initiated the profile compilation phase, logging into the workstation using the enterprise domain account (`MYDOMAIN\Administrator`).

Step 14: Cryptographic Token & Environment Verification

[Image Linked: screenshots/screenshot_14.png]

Technical Context: Executed environmental diagnostics (`whoami`), verifying that the system successfully recognizes the local token under the primary domain administrative security descriptor.